

Reversing $x + W(x)$

Exact r -Lambert Reduction, All-Orders Asymptotics, and a General Calculus for Logarithmic Transseries

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Abstract

Let $W = W_0$ denote the real principal Lambert function and let $f(x) = x + W(x)$ on $[0, \infty)$. We determine the inverse $g = f^{-1}$ at infinity in three mutually compatible forms. First, an exact change of variable shows that

$$g(z) = z - \mathcal{W}_1(z),$$

where $\mathcal{W}_r$ is the r -Lambert function, defined as the inverse of $y \mapsto ye^y + ry$. Second, if $L = W(z)$ and $q = e^{-L} = L/z$, then an analytic reversion problem at the ordinary origin gives the convergent large- z expansion

$$g(z) = z - L + \sum_{n \geq 1} A_n(L) q^n.$$

We give a Lagrange coefficient formula, a triangular recurrence, the first several rational coefficients, a joint expansion in q and $1/L$, and uniform remainder estimates. Third, Lagrange–Bürmann inversion at infinity yields

$$g(z) = z + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dz^{n-1}} W(z)^n,$$

which becomes a rational-coefficient expansion in powers of $1/z$. Substitution of the classical Stirling-number expansion of W produces the full polynomial-logarithmic transseries in $\log z$ and $\log \log z$. A key point is that the correction to $z - W(z)$ starts at order $W(z)/z$, beyond every fixed inverse power of $\log z$.

The second half of the article extracts a reusable method for reversing near-identity logarithmic transseries. It treats slow coefficient fields, valuation gain under differentiation, fixed-point and coefficient recursions, Lagrange–Bürmann inversion, Newton acceleration, domain promotion, residual certification, and the family $x + \alpha W(x)$, as well as the more general perturbation $x + h(W(x))$.

Main result

For $z \rightarrow +\infty$, put $L = W(z)$. Then

$$\begin{aligned} f^{-1}(z) = z - L + & \frac{L^2}{z(1+L)} + \frac{L^3(L^2-2)}{2z^2(1+L)^3} \\ & + \frac{L^4(2L-1)(L^3-6L-6)}{6z^3(1+L)^5} \\ & + \frac{L^5(6L^6-8L^5-69L^4-40L^3+96L^2+72L-24)}{24z^4(1+L)^7} + O\left(\frac{L^5}{z^5}\right). \end{aligned}$$

More generally, after retaining the terms through z^{-N} , the error is

$$f^{-1}(z) - G_N(z) \sim \frac{1}{N+1} \left(\frac{W(z)}{z} \right)^{N+1}.$$

At the coarser inverse-logarithmic scale, with $L_1 = \log z$ and $L_2 = \log \log z$,

$$\begin{aligned} f^{-1}(z) \sim & z - L_1 + L_2 - \frac{L_2}{L_1} + \frac{L_2(2 - L_2)}{2L_1^2} \\ & - \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} + \frac{L_2(12 - 36L_2 + 22L_2^2 - 3L_2^3)}{12L_1^4} + \dots \end{aligned}$$

The first displayed expansion resolves the transseries sectors that the second one cannot see.

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1 Introduction and orientation

The Lambert function is defined by

$$W(x)e^{W(x)} = x. \quad (1.1)$$

On the nonnegative real axis the principal branch is nonnegative, smooth, and increasing. The map

$$f(x) = x + W(x), \quad x \geq 0, \quad (1.2)$$

is therefore strictly increasing from $[0, \infty)$ onto itself, and its inverse $g = f^{-1}$ is globally defined there.

At first sight the inverse appears to be only a routine near-identity reversion, because $W(x) = \mathcal{O}(\log x) = o(x)$. The problem is more instructive than that description suggests. Three asymptotic scales occur:

1. the leading scale z ;
2. the slow logarithmic scale generated by $L_1 = \log z$ and $L_2 = \log L_1$;
3. the much smaller scale $q = W(z)/z = e^{-W(z)}$, whose powers form separate transseries sectors.

A calculation performed only in powers of $1/\log z$ determines $z - W(z)$ to arbitrary inverse-logarithmic order, but it cannot detect the first correction of size $W(z)/z$. This article keeps the scales separate and then explains how to combine them safely.

The linked ProveIt draft collection develops algebra, composition, and reversion for polynomial-logarithmic transseries [9]. The present note applies that general viewpoint to one focused inverse problem and pushes the special structure far enough to obtain an exact ordinary power-series reversion, explicit rational coefficients, and a convergent sector expansion.

1.1 Notation and the three coordinate systems

Throughout, W without a branch label means W_0 on $[0, \infty)$. To avoid confusing a classical branch index with the parameter of a generalized Lambert function, we use the following notation.

Definition 1.1 (The r -Lambert function). For a fixed real parameter r , let $\mathcal{W}_r$ denote a local inverse of

$$y \mapsto ye^y + ry. \quad (1.3)$$

When $r = 1$, this map is strictly increasing on the real line, so $\mathcal{W}_1 : \mathbb{R} \rightarrow \mathbb{R}$ is single-valued. The notation $\mathcal{W}_r$ is used here in place of the common notation W_r , because W_1 also denotes a nonprincipal complex branch of the classical Lambert function. This generalized inverse is the r -Lambert function studied by Mező and Baricz [3].

We shall move among:

$$x \xleftrightarrow{y=W(x)} y, \quad z \xleftrightarrow{L=W(z)} L, \quad q = e^{-L} = \frac{L}{z}.$$

The y -coordinate gives the exact generalized-Lambert reduction. The L -coordinate rationalizes differentiation. The q -coordinate turns the exponentially small correction into an ordinary convergent power series.

2 Exact inversion and global real-variable properties

2.1 Reduction to the r -Lambert function

Theorem 2.1 (Exact inverse). For every $z \geq 0$,

$$g(z) = f^{-1}(z) = z - \mathcal{W}_1(z) = \mathcal{W}_1(z)e^{\mathcal{W}_1(z)}. \quad (2.1)$$

Equivalently, if $y = \mathcal{W}_1(z)$, then $g(z) = ye^y$ and $W(g(z)) = y$.

Proof. Put $y = W(x)$. Then $x = ye^y$, and the equation $z = x + W(x)$ becomes

$$z = ye^y + y.$$

By definition, $y = \mathcal{W}_1(z)$. Since $z = x + y$, we obtain $x = z - y = z - \mathcal{W}_1(z)$. The identity $x = ye^y$ gives the second form in (2.1). All quantities are nonnegative, so the principal real branches are the relevant ones throughout. $\square$

Lean status. Formal, in `LambertShiftInverse.lean`: $\mathcal{W}_1$ is `Fabius.shiftedLambertW` (the inverse on $[0, \infty)$ of `Fabius.shiftedMulExp`, $u \mapsto ue^u + u$), g is `Fabius.lambertShiftInv`, and the three identities are `Fabius.lambertShift_lambertShiftInv` ($f(g(z)) = z$), `Fabius.lambertShiftInv_eq_mul_exp` ($g(z) = ye^y$) and `Fabius.principalLambertW_lambertShiftInv` ($W(g(z)) = y$); g is the two-sided inverse of f on $[0, \infty)$ by `Fabius.lambertShiftInv_lambertShift` and `Fabius.lambertShift_eq_iff`. The bracket (2.3) is `Fabius.one_le_deriv_lambertShift`, `Fabius.deriv_lambertShift_le_two`, and the strict form of (2.6) for $z > 0$ is `Fabius.lambertShiftInv_lt_self`, `Fabius.sub_principalLambertW_lt_lambertShiftInv`.

This exact formula identifies the special function involved, but it does not by itself expose the full large- z hierarchy. The remainder of the article derives that hierarchy directly.

2.2 Monotonicity, derivative bounds, and elementary brackets

Differentiating (1.1) gives

$$W'(x) = \frac{W(x)}{x(1 + W(x))} = \frac{e^{-W(x)}}{1 + W(x)}, \quad x > 0, \quad (2.2)$$

with the continuous value $W'(0) = 1$. Hence

$$1 \leq f'(x) = 1 + W'(x) \leq 2, \quad x \geq 0. \quad (2.3)$$

This proves global invertibility and also yields

$$\frac{1}{2} \leq g'(z) < 1, \quad z \geq 0. \quad (2.4)$$

Since $W''(x) < 0$ for $x > 0$, the function f is concave and its inverse g is convex.

The defining equation can also be written

$$g(z) = z - W(g(z)). \quad (2.5)$$

Because $0 \leq g(z) \leq z$ and W is increasing, we obtain the useful global bracket

$$z - W(z) \leq g(z) \leq z. \quad (2.6)$$

The lower approximation $z - W(z)$ is already correct through every fixed inverse-logarithmic order. Its positive error is nevertheless of order $W(z)/z$, as will be proved exactly below.

Proposition 2.2 (Residual-to-error certificate). *Let $\tilde{g}(z) \geq 0$ be any approximation and set*

$$R(z) = \tilde{g}(z) + W(\tilde{g}(z)) - z. \quad (2.7)$$

Then

$$\frac{|R(z)|}{2} \leq |\tilde{g}(z) - g(z)| \leq |R(z)|. \quad (2.8)$$

Moreover, $R(z)$ and $\tilde{g}(z) - g(z)$ have the same sign.

Proof. Apply the mean-value theorem to $f(\tilde{g}) - f(g) = R$ and use (2.3). $\square$

Lean status. The sandwich (2.8) is formal for every $z \geq 0$ and every $\tilde{g}(z) \geq 0$: `Fabius.abs_sub_lambertShiftInv_le_abs_residual` and `Fabius.abs_residual_le_two_mul_abs_sub_lambertShiftInv` in `LambertShiftInverse.lean`, from the two-sided Lipschitz bracket of f (`Fabius.abs_sub_le_abs_lambertShift_sub`, `Fabius.abs_lambertShift_sub_le_two_mul`), itself the mean-value theorem with (2.3). The sign statement is `Fabius.residual_pos_iff` and `Fabius.residual_neg_iff`.

This elementary certificate will later turn a formal composition check into a rigorous numerical error bound.

3 The natural small parameter and analytic reversion

3.1 Straightening the dominant Lambert behavior

Let

$$L = W(z), \quad z = Le^L, \quad q = e^{-L} = \frac{L}{z}. \quad (3.1)$$

Write the generalized-Lambert solution in the form

$$\mathcal{W}_1(z) = L - v. \quad (3.2)$$

By Theorem 2.1,

$$g(z) = z - L + v. \quad (3.3)$$

Substitution of (3.2) into $(L - v)e^{L-v} + L - v = Le^L$, followed by division by e^L , gives

$$(L - v)(e^{-v} + e^{-L}) = L.$$

Thus the correction v is characterized exactly by

$$q = B_L(v) := \frac{L}{L - v} - e^{-v}. \quad (3.4)$$

The right side vanishes at $v = 0$ and

$$B'_L(0) = 1 + \frac{1}{L} > 0.$$

We have therefore reduced the large-variable problem to ordinary local series inversion at the origin.

It is convenient to set

$$u = \frac{1}{L}. \quad (3.5)$$

Then

$$B(v, u) = \frac{1}{1 - uv} - e^{-v} = \sum_{k \geq 1} b_k(u) v^k, \quad b_k(u) = u^k - \frac{(-1)^k}{k!}. \quad (3.6)$$

In particular, $b_1(u) = 1 + u$.

3.2 A convergent sector expansion

Theorem 3.1 (Joint analytic inverse). *There are neighborhoods of $(q, u) = (0, 0)$ and a unique analytic function $V(q, u)$ such that*

$$B(V(q, u), u) = q, \quad V(0, u) = 0. \quad (3.7)$$

Consequently, for all sufficiently large positive z ,

$$g(z) = z - L + V\left(e^{-L}, \frac{1}{L}\right) = z - L + \sum_{n \geq 1} A_n(L) e^{-nL}, \quad L = W(z), \quad (3.8)$$

where the series converges. Equivalently,

$$g(z) = z - L + \sum_{n \geq 1} A_n(L) \left(\frac{L}{z}\right)^n. \quad (3.9)$$

The coefficients are analytic functions of $u = 1/L$ near $u = 0$.

Proof. The function

$$F(v, q, u) = B(v, u) - q$$

is analytic near $(0, 0, 0)$, satisfies $F(0, 0, 0) = 0$, and has

$$\frac{\partial F}{\partial v}(0, 0, 0) = 1.$$

The analytic implicit-function theorem gives a unique analytic solution $v = V(q, u)$. Along the positive-real path determined by $u = 1/W(z)$ and $q = e^{-W(z)}$, both parameters tend to zero. Hence that path eventually enters the common domain of analyticity. Substituting into (3.3) proves the result. $\square$

This theorem is stronger than a formal asymptotic statement: the outer sector series in powers of $q = L/z$ is genuinely convergent for all sufficiently large z . Each coefficient can then be expanded further in powers of $1/L$, producing the nested logarithmic transseries.

3.3 Lagrange coefficient formula

Since $B_L(v) = b_1 v + \mathcal{O}(v^2)$, ordinary Lagrange inversion gives the following all-orders expression.

Theorem 3.2 (Coefficient formula). *For $n \geq 1$,*

$$A_n(L) = \frac{1}{n} [v^{n-1}] \left(\frac{v}{B_L(v)} \right)^n \quad (3.10)$$

$$= \frac{1}{n!} \frac{d^{n-1}}{dv^{n-1}} \left(\frac{v}{\frac{L}{L-v} - e^{-v}} \right)^n \Big|_{v=0}. \quad (3.11)$$

Equivalently, in the variable $u = 1/L$, replace $B_L(v)$ by $B(v, u)$ from (3.6).

Proof. Apply the standard Lagrange inversion formula to $q = B_L(v)$, or to $v = q\phi(v)$ with $\phi(v) = v/B_L(v)$; see, for example, Comtet [4] or Flajolet and Sedgewick [5]. $\square$

3.4 Triangular coefficient recursion

The coefficient formula is compact, but a triangular recurrence is often more efficient in exact symbolic arithmetic. Write

$$V(q, u) = \sum_{n \geq 1} A_n(u) q^n, \quad A_n(u) = A_n(1/u). \quad (3.12)$$

Substituting into $q = \sum_{k \geq 1} b_k(u) V^k$ gives

$$A_1(u) = \frac{1}{1+u}, \quad (3.13)$$

and, for $n \geq 2$,

$$A_n(u) = -\frac{1}{1+u} \sum_{k=2}^n b_k(u) \sum_{\substack{j_1+\dots+j_k=n \\ j_i \geq 1}} A_{j_1}(u) \cdots A_{j_k}(u). \quad (3.14)$$

The occurrence of A_n is linear and confined to the term $b_1 A_n$; all other entries have lower indices. This is the fundamental triangularity behind near-identity reversion.

Proposition 3.3 (Rational shape of the coefficients). *For each $n \geq 1$, there is a polynomial $R_n(L) \in \mathbb{Z}[L]$ of degree $2n - 2$ such that*

$$A_n(L) = \frac{LR_n(L)}{n!(1+L)^{2n-1}}. \quad (3.15)$$

The leading coefficient of R_n is $(n-1)!$. Consequently,

$$A_n(L) \longrightarrow \frac{1}{n} \quad (L \rightarrow +\infty). \quad (3.16)$$

Proof. Factor the linear coefficient from $B(v, u)$:

$$B(v, u) = (1+u)v \left(1 + \sum_{j \geq 1} \frac{b_{j+1}(u)}{1+u} v^j \right).$$

The coefficient form of Lagrange inversion therefore gives

$$\begin{aligned} A_n(u) &= \frac{1}{n(1+u)^n} [v^{n-1}] \left(1 + \sum_{j \geq 1} \frac{b_{j+1}(u)}{1+u} v^j \right)^{-n} \\ &= \frac{1}{n(1+u)^n} \sum_{m=0}^{n-1} (-1)^m \binom{n+m-1}{m} \frac{1}{(1+u)^m} [v^{n-1}] \left(\sum_{j \geq 1} b_{j+1}(u) v^j \right)^m. \end{aligned}$$

After passage to the common denominator $(1+u)^{2n-1}$, every term has numerator degree at most

$$(n-1-m) + \sum_{i=1}^m (j_i + 1) = (n-1-m) + (n-1+m) = 2n-2,$$

where $j_1 + \dots + j_m = n-1$. It remains to check the asserted integrality after multiplication by $n!$. Group the ordered compositions by multiplicities c_j , where $\sum_j c_j = m$ and $\sum_j j c_j = n-1$. Choosing from each $b_{j+1}(u) = u^{j+1} - (-1)^{j+1}/(j+1)!$ either its monomial part or its constant part reduces the possible denominators to products of selected factorials $(j+1)!$. The potentially largest denominator is cleared by

$$(n-1)! \binom{n+m-1}{m} \frac{m!}{\prod_j c_j!} \prod_j \frac{1}{((j+1)!)^{c_j}} = \frac{(n+m-1)!}{\prod_j c_j! ((j+1)!)^{c_j}},$$

which is an integer: it counts set partitions of an $(n+m-1)$ -element labelled set into c_j unlabeled blocks of size $j+1$. This is the case in which every factor contributes its constant part, and therefore it has the largest possible factorial denominator. If, for a given j , only $d_j \leq c_j$ factors contribute their

constant parts, the grouped coefficient is the preceding integer multiplied by $\binom{c_j}{d_j}((j+1)!)^{c_j-d_j}$, and similarly for all j . It is therefore integral as well. Hence

$$A_n(u) = \frac{S_n(u)}{n!(1+u)^{2n-1}}$$

for a polynomial $S_n \in \mathbb{Z}[u]$ of degree at most $2n-2$. Substituting $u = 1/L$ yields

$$A_n(1/L) = \frac{L R_n(L)}{n!(1+L)^{2n-1}}, \quad R_n(L) = L^{2n-2} S_n(1/L) \in \mathbb{Z}[L].$$

Finally, at $u = 0$ the inverse equation is

$$q = 1 - e^{-v}, \quad v = -\log(1-q) = \sum_{n \geq 1} \frac{q^n}{n},$$

so $A_n(0) = 1/n$, equivalently $S_n(0) = (n-1)!$. Thus R_n has degree exactly $2n-2$ and leading coefficient $(n-1)!$, completing the proof. $\square$

3.5 The first coefficients

The first five coefficients are

$$A_1(L) = \frac{L}{L+1}, \tag{3.17}$$

$$A_2(L) = \frac{L(L^2-2)}{2(L+1)^3}, \tag{3.18}$$

$$A_3(L) = \frac{L(2L-1)(L^3-6L-6)}{6(L+1)^5}, \tag{3.19}$$

$$A_4(L) = \frac{L}{24(L+1)^7} (6L^6 - 8L^5 - 69L^4 - 40L^3 + 96L^2 + 72L - 24), \tag{3.20}$$

$$A_5(L) = \frac{L}{120(L+1)^9} (24L^8 - 58L^7 - 428L^6 - 151L^5 + 1320L^4 + 1380L^3 - 480L^2 - 720L + 120). \tag{3.21}$$

They can be generated either from (3.11), from (3.14), or by the differential-operator method developed in the next section.

3.6 Resumming all q -sectors at fixed inverse powers of L

The joint analyticity in (q, u) permits the opposite organization: first expand in $u = 1/L$, while retaining the dependence on q exactly. Put

$$s = 1 - q, \quad t = -\log(1 - q). \tag{3.22}$$

Solving (3.6) coefficient by coefficient in u gives

$$V(q, u) = t - \frac{ut}{s} + \frac{u^2 t (2 + (2q-1)t)}{2s^2} + O(u^3), \tag{3.23}$$

locally uniformly for q in a compact subset of $|q| < 1$. Thus a useful diagonally resummed approximation is

$$g(z) = z - L - \log(1 - q) + \frac{\log(1 - q)}{L(1 - q)} + O\left(\frac{-\log(1 - q)}{L^2(1 - q)^2}\right), \quad q = \frac{L}{z}. \tag{3.24}$$

At leading order in $1/L$, all sector coefficients $1/n$ sum to $-\log(1 - q)$. The ordinary expansion $\sum_n A_n(L) q^n$, however, is usually the cleaner form for high-order exact computation.

4 Lagrange–Bürmann inversion at infinity

The preceding section exploits a special exact coordinate. A second derivation starts directly from the near-identity map

$$f(x) = x + H(x), \quad H(x) = W(x),$$

and is the prototype for general logarithmic transseries reversal.

4.1 The universal near-identity formula

Theorem 4.1 (Lagrange–Bürmann formula at infinity). *Formally, and asymptotically as $z \rightarrow +\infty$,*

$$g(z) = z + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dz^{n-1}} (W(z)^n). \quad (4.1)$$

For every fixed power z^{-N} , only finitely many summands contribute. In the present problem the resulting expansion agrees term by term with the convergent sector expansion (3.9).

Proof. Introduce a formal parameter ε and solve

$$K(\varepsilon) = -\varepsilon H(z + K(\varepsilon)), \quad K(0) = 0.$$

Ordinary Lagrange inversion in ε , with coefficients in the differential algebra of functions of z , gives

$$[\varepsilon^n]K(\varepsilon) = \frac{(-1)^n}{n!} D^{n-1}(H^n), \quad D = \frac{d}{dz}.$$

Setting $\varepsilon = 1$ yields (4.1). Here each differentiation contributes one additional power of $1/z$ in the W -coordinate algebra, so the family is summable coefficientwise in $1/z$. The sum satisfies the fixed-point equation $K = -H(z + K)$, hence is the inverse correction. Agreement with (3.9) follows from uniqueness of the formal inverse, or directly from Theorem 4.3 below. $\square$

Expanding the first few universal differential polynomials gives

$$\begin{aligned} (z + H)^{(-1)} &= z - H + HH' - H(H')^2 - \frac{1}{2}H^2H'' \\ &\quad + H(H')^3 + \frac{3}{2}H^2H'H'' + \frac{1}{6}H^3H''' + \dots \end{aligned} \quad (4.2)$$

For a short perturbation H , this formula often produces the first blocks faster than direct composition.

4.2 An exact differential operator in the W -coordinate

Because $L = W(z)$,

$$\frac{dL}{dz} = \frac{L}{z(1+L)}. \quad (4.3)$$

For a rational function $R(L)$, define

$$\mathcal{D}_k R := \frac{L}{1+L} \frac{dR}{dL} - kR. \quad (4.4)$$

Then

$$\frac{d}{dz} \left(z^{-k} R(L) \right) = z^{-k-1} \mathcal{D}_k R(L). \quad (4.5)$$

Thus differentiation raises the primary $1/z$ -valuation by exactly one, while all slow dependence remains in a rational coefficient.

Theorem 4.2 (Coefficient operator formula). *Define $C_n(L)$ for $n \geq 1$ by*

$$C_n(L) = \frac{(-1)^{n+1}}{(n+1)!} (\mathcal{D}_{n-1} \mathcal{D}_{n-2} \cdots \mathcal{D}_1 \mathcal{D}_0) L^{n+1}. \quad (4.6)$$

Then

$$g(z) = z - L + \sum_{n \geq 1} \frac{C_n(L)}{z^n}, \quad L = W(z). \quad (4.7)$$

Furthermore,

$$C_n(L) = \frac{L^{n+1} R_n(L)}{n!(1+L)^{2n-1}}, \quad (4.8)$$

where the polynomial R_n is the same as in (3.15).

Proof. The term with index $n+1$ in (4.1) is

$$\frac{(-1)^{n+1}}{(n+1)!} D^n(L^{n+1}).$$

Repeated use of (4.5), beginning at $k=0$, gives (4.6) and shows that this term is exactly $z^{-n} C_n(L)$. On the other hand, the convergent sector expansion (3.9) writes the same inverse as

$$z - L + \sum_{n \geq 1} z^{-n} L^n A_n(L).$$

Uniqueness in the formal coefficient algebra $\mathbb{Q}(L)[[z^{-1}]]$ therefore gives $C_n(L) = L^n A_n(L)$. Substitution of (3.15) proves (4.8) with the same polynomial R_n . $\square$

Proposition 4.3 (Equivalence of the two coefficient systems). *For every $n \geq 1$,*

$$C_n(L) = L^n A_n(L). \quad (4.9)$$

Consequently,

$$\frac{C_n(L)}{z^n} = A_n(L) \left(\frac{L}{z} \right)^n.$$

Proof. Both (3.9) and (4.7) are expansions of the unique inverse in powers of $q = L/z$. Comparing the coefficient of q^n gives the identity. It can also be verified immediately by combining (3.15) and (4.8). $\square$

4.3 Explicit rational-coefficient expansion

The first five C_n are

$$C_1(L) = \frac{L^2}{L+1}, \quad (4.10)$$

$$C_2(L) = \frac{L^3(L^2-2)}{2(L+1)^3}, \quad (4.11)$$

$$C_3(L) = \frac{L^4(2L-1)(L^3-6L-6)}{6(L+1)^5}, \quad (4.12)$$

$$C_4(L) = \frac{L^5}{24(L+1)^7} (6L^6 - 8L^5 - 69L^4 - 40L^3 + 96L^2 + 72L - 24), \quad (4.13)$$

$$C_5(L) = \frac{L^6}{120(L+1)^9} (24L^8 - 58L^7 - 428L^6 - 151L^5 + 1320L^4 + 1380L^3 - 480L^2 - 720L + 120). \quad (4.14)$$

Therefore

$$\begin{aligned}
g(z) = & z - L + \frac{L^2}{z(L+1)} + \frac{L^3(L^2-2)}{2z^2(L+1)^3} \\
& + \frac{L^4(2L-1)(L^3-6L-6)}{6z^3(L+1)^5} \\
& + \frac{L^5(6L^6-8L^5-69L^4-40L^3+96L^2+72L-24)}{24z^4(L+1)^7} \\
& + \frac{L^6(24L^8-58L^7-428L^6-151L^5+1320L^4+1380L^3-480L^2-720L+120)}{120z^5(L+1)^9} \\
& + O\left(\frac{L^6}{z^6}\right), \quad L = W(z).
\end{aligned} \tag{4.15}$$

The remainder notation in (4.15) is justified by the convergent q -series and the boundedness of $A_n(L)$ for fixed n as $L \rightarrow \infty$.

4.4 Large- L structure of the sector coefficients

The leading behavior $A_n(L) \rightarrow 1/n$ implies

$$C_n(L) \sim \frac{L^n}{n}. \tag{4.16}$$

For example,

$$C_1(L) = L - 1 + \frac{1}{L} - \frac{1}{L^2} + \frac{1}{L^3} + O(L^{-4}), \tag{4.17}$$

$$C_2(L) = \frac{1}{2}L^2 - \frac{3}{2}L + 2 - \frac{2}{L} + \frac{3}{2L^2} + O(L^{-3}), \tag{4.18}$$

$$C_3(L) = \frac{1}{3}L^3 - \frac{11}{6}L^2 + \frac{23}{6}L - \frac{31}{6} + \frac{31}{6L} + O(L^{-2}). \tag{4.19}$$

Thus the n -th outer sector begins with $(L/z)^n/n$, and summing only these leading diagonal terms recovers the $-\log(1-q)$ resummation in (3.23).

5 Expansion in $\log z$ and $\log \log z$

5.1 The classical Lambert expansion

Set

$$L_1 = \log z, \quad L_2 = \log L_1. \tag{5.1}$$

Let $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$ denote an unsigned Stirling cycle number of the first kind, and define

$$P_n(s) = \sum_{m=1}^n (-1)^{n-m} \left[\begin{smallmatrix} n \\ n-m+1 \end{smallmatrix} \right] \frac{s^m}{m!}. \tag{5.2}$$

The standard large-argument expansion of the principal Lambert function is

$$W(z) \sim L_1 - L_2 + \sum_{n \geq 1} \frac{P_n(L_2)}{L_1^n}. \tag{5.3}$$

The first terms are

$$\begin{aligned}
 W(z) \sim & L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(L_2 - 2)}{2L_1^2} + \frac{L_2(2L_2^2 - 9L_2 + 6)}{6L_1^3} \\
 & + \frac{L_2(3L_2^3 - 22L_2^2 + 36L_2 - 12)}{12L_1^4} \\
 & + \frac{L_2(12L_2^4 - 125L_2^3 + 350L_2^2 - 300L_2 + 60)}{60L_1^5} + \dots .
 \end{aligned} \tag{5.4}$$

This expansion is asymptotic in the Poincaré sense on the positive real axis. It is also obtainable by an ordinary convergent double-series reversion in the variables $1/L_1$ and L_2/L_1 ; see Corless et al. [1] and DLMF §4.13 [2].

5.2 The ordinary inverse-logarithmic sector

Since $g(z) = z - W(z) + O(W(z)/z)$, the correction sectors in powers of $1/z$ are smaller than every fixed inverse power of L_1 . Thus, for every fixed N ,

$$g(z) = z - L_1 + L_2 - \sum_{n=1}^N \frac{P_n(L_2)}{L_1^n} + O\left(\frac{L_2^{N+1}}{L_1^{N+1}}\right). \tag{5.5}$$

In explicit form,

$$\begin{aligned}
 g(z) \sim & z - L_1 + L_2 - \frac{L_2}{L_1} + \frac{L_2(2 - L_2)}{2L_1^2} \\
 & - \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} + \frac{L_2(12 - 36L_2 + 22L_2^2 - 3L_2^3)}{12L_1^4} \\
 & - \frac{L_2(12L_2^4 - 125L_2^3 + 350L_2^2 - 300L_2 + 60)}{60L_1^5} + \dots .
 \end{aligned} \tag{5.6}$$

Equation (5.6) is a valid answer at the ordinary logarithmic scale. It is not the complete transseries: any finite or infinite formal manipulation confined to the z^0 inverse-logarithmic sector misses terms of order L_1/z , L_1^2/z^2 , and so on.

5.3 Why the first new sector is beyond all inverse-logarithmic orders

For each fixed M ,

$$\frac{L_1/z}{L_1^{-M}} = \frac{L_1^{M+1}}{z} \longrightarrow 0. \tag{5.7}$$

Hence L_1/z is beyond all algebraic orders in $1/L_1$. The first exact correction to $z - W(z)$ is

$$\frac{W(z)^2}{z(1 + W(z))} \sim \frac{\log z}{z}. \tag{5.8}$$

This explains a common apparent paradox: two approximations can agree to all displayed inverse-logarithmic orders and still differ by a structured, computable transseries sector.

A direct residual calculation makes the mechanism visible. Let $G_0 = z - L$. Then

$$f(G_0) - z = W(z - L) - L.$$

Taylor expansion around z gives

$$W(z - L) - L = -LW'(z) + O(L^2W''(z)) = -\frac{L^2}{z(1+L)} + O\left(\frac{L^2}{z^2}\right).$$

Since $f' \sim 1$, the correction needed to cancel this residual is precisely the first term in (4.15).

5.4 A rectangular polynomial-logarithmic transseries

One may expand each rational function $C_n(W(z))$ in L_1^{-1} after fixing the outer sector z^{-n} . For example,

$$\begin{aligned} C_1(W(z)) = L_1 - L_2 - 1 + \frac{L_2 + 1}{L_1} + \frac{L_2^2/2 - 1}{L_1^2} \\ + \frac{L_2^3/3 - L_2^2/2 - 2L_2 + 1}{L_1^3} + O\left(\frac{L_2^4}{L_1^4}\right), \end{aligned} \quad (5.9)$$

$$\begin{aligned} C_2(W(z)) = \frac{1}{2}L_1^2 - \left(L_2 + \frac{3}{2}\right)L_1 + \frac{1}{2}L_2^2 + \frac{5}{2}L_2 + 2 \\ - \frac{L_2^2/2 + 5L_2/2 + 2}{L_1} + O\left(\frac{L_2^3}{L_1^2}\right), \end{aligned} \quad (5.10)$$

$$C_3(W(z)) = \frac{1}{3}L_1^3 - \left(L_2 + \frac{11}{6}\right)L_1^2 + \left(L_2^2 + \frac{14}{3}L_2 + \frac{23}{6}\right)L_1 + O(L_2^3). \quad (5.11)$$

Accordingly, in lexicographic transseries order,

$$\begin{aligned} g(z) \sim_{\text{tr}} z - L_1 + L_2 - \frac{L_2}{L_1} + \frac{L_2(2 - L_2)}{2L_1^2} - \dots \\ + \frac{1}{z} \left\{ L_1 - L_2 - 1 + \frac{L_2 + 1}{L_1} + \frac{L_2^2/2 - 1}{L_1^2} + \dots \right\} \\ + \frac{1}{z^2} \left\{ \frac{1}{2}L_1^2 - \left(L_2 + \frac{3}{2}\right)L_1 + \frac{1}{2}L_2^2 + \frac{5}{2}L_2 + 2 + \dots \right\} \\ + \frac{1}{z^3} \left\{ \frac{1}{3}L_1^3 - \left(L_2 + \frac{11}{6}\right)L_1^2 + \left(L_2^2 + \frac{14}{3}L_2 + \frac{23}{6}\right)L_1 + \dots \right\} + \dots \end{aligned} \quad (5.12)$$

The symbol $\sim_{\text{tr}}$ emphasizes that the ordering is hierarchical: first by the outer power z^{-n} , then by the inverse-log order within its coefficient. A finite numerical approximation should specify both cutoffs.

6 Remainders, convergence, and certification

6.1 Uniform sector remainders

Let

$$G_N(z) = z - L + \sum_{n=1}^N A_n(L)q^n, \quad L = W(z), \quad q = e^{-L}. \quad (6.1)$$

The analytic inverse from [Theorem 3.1](#) immediately gives a uniform remainder theorem.

Theorem 6.1 (Sector remainder). *For every fixed $N \geq 0$,*

$$g(z) - G_N(z) = O(q^{N+1}) = O\left(\frac{W(z)^{N+1}}{z^{N+1}}\right), \quad z \rightarrow +\infty. \quad (6.2)$$

More precisely,

$$g(z) - G_N(z) \sim \frac{q^{N+1}}{N+1} = \frac{1}{N+1} \left(\frac{W(z)}{z} \right)^{N+1}. \quad (6.3)$$

In particular, the truncation error is positive for all sufficiently large z .

Proof. The convergent Taylor expansion of $V(q, u)$ gives

$$V(q, u) - \sum_{n=1}^N A_n(u) q^n = A_{N+1}(u) q^{N+1} + O(q^{N+2})$$

uniformly for u near zero. By (3.16), $A_{N+1}(u) \rightarrow 1/(N+1)$. Substitution of $u = 1/L$ and $q = L/z$ proves both claims. $\square$

A quantitative Cauchy estimate is also available. Choose a closed bidisc $|q| \leq \rho$, $|u| \leq \eta$ lying inside the analytic domain of V , and let $M = \max |V|$ on its distinguished boundary. For $|q| < \rho$,

$$\left| V(q, u) - \sum_{n=1}^N A_n(u) q^n \right| \leq \frac{M(|q|/\rho)^{N+1}}{1 - |q|/\rho}. \quad (6.4)$$

This bound is conservative but completely explicit once a working bidisc is chosen.

6.2 Composition residuals

A symbolic system need not know the analytic constants in (6.4). It can instead compute

$$R_N(z) = G_N(z) + W(G_N(z)) - z. \quad (6.5)$$

Formal cancellation through the retained sector gives

$$R_N(z) = O(q^{N+1}).$$

The residual certificate [Theorem 2.2](#) then supplies an actual error enclosure:

$$\frac{|R_N(z)|}{2} \leq |G_N(z) - g(z)| \leq |R_N(z)|. \quad (6.6)$$

This is particularly valuable for computer algebra: construct the series exactly, evaluate it numerically, evaluate one Lambert W , and obtain a rigorous a posteriori error bound without evaluating W_1 .

6.3 Differentiated asymptotics

The exact derivative identity

$$g'(z) = \frac{1}{1 + W'(g(z))} \quad (6.7)$$

provides an independent check. Differentiating any truncation of (4.7) term by term and substituting it into (6.7) must leave a residual beyond the target order. Because differentiation raises the $1/z$ -valuation in the W -coordinate field, the differentiated transseries is as well ordered as the original one.

7 Numerical verification

For reference, the exact value in this section is computed by solving

$$ye^y + y = z \quad (7.1)$$

for its unique nonnegative root and then setting $g(z) = z - y$. Let G_N be the approximation (6.1); thus $N = 0$ means $G_0 = z - W(z)$.

Table 1: Absolute error $|G_N(z) - g(z)|$.

z	$N = 0$	$N = 1$	$N = 2$	$N = 3$	$N = 4$	$N = 5$
10	1.12×10^{-1}	1.05×10^{-3}	2.99×10^{-4}	2.34×10^{-5}	6.89×10^{-7}	2.51×10^{-7}
100	2.64×10^{-2}	2.19×10^{-4}	9.50×10^{-7}	2.35×10^{-8}	7.95×10^{-10}	1.25×10^{-11}
1000	4.42×10^{-3}	7.59×10^{-6}	1.35×10^{-8}	1.72×10^{-11}	1.10×10^{-14}	1.81×10^{-16}
10^6	1.05×10^{-5}	4.96×10^{-11}	2.93×10^{-16}	1.84×10^{-21}	1.16×10^{-26}	7.16×10^{-32}

The corresponding exact values are

$$\begin{aligned} g(10) &= 8.366493829844153615806835 \dots, \\ g(100) &= 96.64072495463040645894733 \dots, \\ g(1000) &= 994.7548143481392806156898 \dots, \\ g(10^6) &= 999988.6166523780211244849 \dots \end{aligned}$$

The data show the sector nature clearly. Once z is moderately large, each added block contributes roughly another factor $q = W(z)/z$, in accord with (6.3). At $z = 10$, where q is not extremely small, the first several terms still improve the approximation, although the early coefficients have not yet settled near their limiting values $1/n$.

8 A general method for logarithmic-transseries reversal

The special calculation above is an instance of a general and highly algorithmic principle. This section develops that principle in a form suited to polynomial-logarithmic transseries at infinity.

8.1 Slow coefficient fields and the primary valuation

Let $X \rightarrow +\infty$ and put $t = X^{-1}$. Introduce iterated logarithms

$$\ell_1 = \log X, \quad \ell_{j+1} = \log \ell_j. \quad (8.1)$$

A depth- d slow coefficient field $\mathcal{C}_d$ may be taken to contain rational functions and asymptotic Laurent series in these variables. The precise Hahn-field completion can be chosen to match the application; a systematic framework is developed by van der Hoeven [8]. What matters for near-identity reversion here is closure under the slow derivation

$$\delta = \frac{\partial}{\partial \ell_1} + \frac{1}{\ell_1} \frac{\partial}{\partial \ell_2} + \frac{1}{\ell_1 \ell_2} \frac{\partial}{\partial \ell_3} + \dots + \frac{1}{\ell_1 \dots \ell_{d-1}} \frac{\partial}{\partial \ell_d}. \quad (8.2)$$

Indeed, for $a \in \mathcal{C}_d$,

$$\frac{da}{dX} = t \delta a. \quad (8.3)$$

Define the outer block algebra

$$\mathcal{T}_d = \mathcal{C}_d[[t]]. \quad (8.4)$$

If $a \in \mathcal{C}_d$, then

$$\frac{d}{dX}(t^n a) = t^{n+1}(\delta a - na). \quad (8.5)$$

Thus differentiation raises the primary t -valuation by one. This single property is the engine behind composition, fixed-point contraction, and Lagrange summability.

The W -coordinate used earlier is an exact alternative slow field. With $L = W(X)$, take $\mathcal{C} = \mathbb{Q}(L)$ and

$$\delta_W = \frac{L}{1+L} \frac{d}{dL}. \quad (8.6)$$

Then (8.5) becomes exactly (4.5). Choosing an adapted slow coordinate can turn complicated logarithmic differentiation into rational algebra.

8.2 Formal composition near the identity

Let

$$F(X) = X + H(X), \quad H \in \mathcal{T}_d. \quad (8.7)$$

Seek an inverse

$$G(X) = X + K(X), \quad K \in \mathcal{T}_d. \quad (8.8)$$

Taylor composition is defined by

$$H(X + K) = \sum_{j \geq 0} \frac{H^{(j)}(X)}{j!} K^j. \quad (8.9)$$

Because $H^{(j)} \in t^j \mathcal{T}_d$, only finitely many Taylor terms can contribute to any prescribed outer order. Thus the composition is well-defined in the t -adic topology even when K begins at outer order zero.

Theorem 8.1 (Near-identity inverse theorem). *For every $F = X + H$ with $H \in \mathcal{T}_d$, there is a unique $G = X + K$, $K \in \mathcal{T}_d$, such that*

$$F \circ G = G \circ F = X. \quad (8.10)$$

The correction is the unique fixed point of

$$K = -H(X + K). \quad (8.11)$$

Proof. Let $\Phi(K) = -H(X + K)$. For two corrections K_1, K_2 , Taylor's difference formula gives

$$H(X + K_1) - H(X + K_2) = \sum_{j \geq 1} \frac{H^{(j)}(X + K_2)}{j!} (K_1 - K_2)^j.$$

The first derivative lies in $t\mathcal{T}_d$, and the j -th derivative lies in $t^j \mathcal{T}_d$. Therefore

$$v_t(\Phi(K_1) - \Phi(K_2)) \geq v_t(K_1 - K_2) + 1. \quad (8.12)$$

Starting with $K_0 = 0$ and iterating $K_{m+1} = \Phi(K_m)$ produces a Cauchy sequence in the complete t -adic algebra. Its limit is the unique right inverse correction. A right inverse tangent to the identity is also a left inverse: if $J = G \circ F = X + R$, then associativity gives $J \circ J = J$. If $R \neq 0$, the leading nonzero block of $J \circ J - X$ is twice that of R , a contradiction in characteristic zero. Hence $R = 0$. $\square$

8.3 General Lagrange–Bürmann formula

Theorem 8.2 (Formal reversal formula). *Under the hypotheses of Theorem 8.1,*

$$F^{(-1)}(X) = X + \sum_{n \geq 1} \frac{(-1)^n}{n!} D^{n-1}(H(X)^n), \quad D = \frac{d}{dX}. \quad (8.13)$$

The family is summable in the primary t -valuation.

Proof. Repeat the auxiliary-parameter proof of Theorem 4.1. Since H^n has nonnegative primary valuation and D^{n-1} raises that valuation by at least $n - 1$, the n -th summand begins at order t^{n-1} . Therefore only finitely many summands affect a fixed block, and setting the auxiliary parameter to one is formally legitimate. The sum satisfies the fixed-point equation and is identified by uniqueness. $\square$

This is the inverse-at-infinity form of the classical Lagrange–Bürmann formula; standard analytic treatments may be found in Henrici [6], while the formal coefficientwise version used here is developed in the linked ProveIt drafts [9].

Corollary 8.3 (Universal first blocks). *The inverse begins as*

$$\begin{aligned} F^{(-1)} = X - H + HH' - H(H')^2 - \frac{1}{2}H^2H'' \\ + H(H')^3 + \frac{3}{2}H^2H'H'' + \frac{1}{6}H^3H''' + \dots \end{aligned} \quad (8.14)$$

8.4 Direct coefficient recursion

For symbolic work with many blocks, it is often better not to expand the powers H^n . Suppose an approximation G_N has been determined through outer order t^N . Introduce one unknown block $a(\ell_1, \dots, \ell_d)t^{N+1}$, compose, and set the first residual block to zero. Near the identity the new block appears linearly with coefficient one, so the equation is triangular.

Coefficient-by-coefficient reversal

1. Choose the transmonomial hierarchy and a coefficient field closed under the needed algebra and derivation.
2. Normalize the dominant part of F . If it is not X , invert the leading monomial first.
3. Insert $G_N = X + K_N$ into $F \circ G_N - X$, retaining only the first unresolved block.
4. Add one unknown block to K_N , solve the resulting linear coefficient equation, and truncate immediately.
5. Continue to the target order, then check both $F \circ G_N - X$ and $G_N \circ F - X$.

In the special q -reversion of Equation (3.4), this procedure is exactly the recurrence (3.14).

8.5 Newton acceleration

If G is an approximate inverse and

$$R = F \circ G - X, \quad (8.15)$$

linearization gives the formal Newton update

$$G_{\text{new}} = G - \frac{F \circ G - X}{F' \circ G}. \quad (8.16)$$

Because $F' = 1 + t\mathcal{T}_d$, the denominator is a unit. After the scale and branch have been normalized correctly, Newton iteration approximately doubles the number of valid outer blocks at each step. A robust implementation computes two or three blocks by the triangular method, then switches to Newton.

8.6 From a formal inverse to an actual asymptotic inverse

Formal correctness becomes analytic correctness under mild stability conditions; this residual-based passage is compatible with the standard Poincaré asymptotic viewpoint [7]. Suppose an actual function F admits the differentiated transseries used in the calculation and that $F'(x) \geq c > 0$ eventually. If a truncation G_N has composition residual

$$F(G_N(X)) - X = O(M_N(X)), \quad (8.17)$$

then the mean-value theorem gives

$$G_N(X) - F^{-1}(X) = O(M_N(X)). \quad (8.18)$$

Thus residual control is the natural bridge between formal transseries and Poincaré asymptotics. In the present problem $c = 1$, and [Theorem 2.2](#) gives a two-sided numerical enclosure.

9 Scale selection, domain promotion, and common failure modes

The formal theorem assumes that the leading map is the identity and that all coefficients live in a differential field stable under the required operations. Most practical failures of transseries reversal occur before the coefficient recursion begins: the dominant scale is normalized incorrectly, an iterated logarithm is omitted, or a composition leaves the chosen coefficient domain.

9.1 Invert the dominant monomial first

Suppose

$$Y \sim cX^a(\log X)^b, \quad ca \neq 0. \quad (9.1)$$

If $b = 0$, the leading inverse is $(Y/c)^{1/a}$. If $b \neq 0$, put $s = \log X$. Then

$$\left(\frac{Y}{c}\right)^{1/b} = se^{(a/b)s},$$

and therefore

$$X = \exp\left[\frac{b}{a}W_k\left(\frac{a}{b}\left(\frac{Y}{c}\right)^{1/b}\right)\right]. \quad (9.2)$$

The branch k is selected by the real interval or complex sector under study. Expansion of W_k introduces $\log Y$ and $\log \log Y$; thus a second logarithmic level is forced whenever the leading monomial contains a nonzero power of $\log X$.

After the leading map F_0 is inverted, factor

$$F = F_0 \circ \Phi, \quad \Phi = X + \text{smaller terms}. \quad (9.3)$$

Then

$$F^{(-1)} = \Phi^{(-1)} \circ F_0^{(-1)}, \quad (9.4)$$

and the near-identity machinery applies only to Φ .

9.2 Generator-first composition

Before expanding an outer transseries at an inner argument G , compute the images of its generators. For example, if

$$G(X) = cX^\beta (\log X)^\gamma (1 + U(X)), \quad U \prec 1, \quad (9.5)$$

then

$$G^{-1} = c^{-1} X^{-\beta} (\log X)^{-\gamma} (1 + U)^{-1}, \quad (9.6)$$

$$\log G = \beta \log X + \gamma \log \log X + \log c + \log(1 + U). \quad (9.7)$$

The term $\gamma \log \log X$ is not optional. If the coefficient domain contains only polynomials in $\log X$, composition with $(\log X)^\gamma$ generally forces promotion to a depth-two logarithmic field.

9.3 A practical promotion chain

A safe symbolic implementation can use the explicit chain

$$\mathbb{Q}[\ell_1] \subset \mathbb{Q}(\ell_1) \subset \mathbb{Q}((\ell_1^{-1})) \subset \mathbb{Q}((\ell_1^{-1}))[\ell_2] \subset \dots \quad (9.8)$$

Promotion is triggered by a concrete operation:

- division by a nonconstant polynomial coefficient requires a rational function field;
- expansion for large ℓ_1 requires Laurent series in ℓ_1^{-1} ;
- inversion of a leading factor containing $(\log X)^b$ introduces $\log \log X$;
- composition with more complicated leading monomials may introduce additional iterated logarithms or real powers.

Every promotion changes the support order and the meaning of truncation. It should be recorded as part of the result, not hidden as a simplification.

9.4 Failure modes

The following errors are especially common.

Using only the inverse-logarithmic sector.

This gives $z - W(z)$ correctly to arbitrary inverse-log order but misses the $W(z)/z$ sectors. The remedy is to introduce the outer parameter $q = W(z)/z$ or work in $\mathcal{C}[[z^{-1}]]$.

Expanding before choosing the correct slow coordinate.

An expression that is rational in $L = W(z)$ can become very large after premature substitution of the (L_1, L_2) series. Compute exact outer coefficients in L first and expand them in inverse logarithms only at the end.

Ignoring branch information.

The positive real problem has no branch ambiguity, but complex extensions do. Classical branch indices, generalized-Lambert parameters, and chosen logarithm branches must be kept distinct.

Truncating only at the end.

Taylor composition, powers, and rational simplification can create severe expression swell. Truncate after every algebraic operation and cache repeated powers.

Checking only the displayed coefficients.

A plausible pattern is not a proof. Verify a left composition residual, a right composition residual, and, when possible, the differentiated inverse identity.

10 Parametric and functional extensions

10.1 The family $x + \alpha W(x)$

Let

$$f_\alpha(x) = x + \alpha W(x), \quad \alpha \geq 0. \quad (10.1)$$

With $y = W(x)$, the inversion equation is

$$ye^y + \alpha y = z. \quad (10.2)$$

Thus the exact inverse is

$$f_\alpha^{-1}(z) = z - \alpha W_\alpha(z). \quad (10.3)$$

Set again $L = W(z)$, $q = e^{-L}$, and write $W_\alpha(z) = L - v$. The same calculation as before gives

$$B_L(v) = \alpha q. \quad (10.4)$$

Therefore the coefficient functions $A_n(L)$ are universal:

$$f_\alpha^{-1}(z) = z - \alpha L + \alpha \sum_{n \geq 1} A_n(L) (\alpha q)^n. \quad (10.5)$$

Equivalently,

$$f_\alpha^{-1}(z) = z - \alpha L + \sum_{n \geq 1} \alpha^{n+1} \frac{C_n(L)}{z^n}. \quad (10.6)$$

For every fixed N , the remainder after $n = N$ is

$$\sim \frac{\alpha^{N+2}}{N+1} \left(\frac{L}{z} \right)^{N+1}. \quad (10.7)$$

The original problem is the case $\alpha = 1$.

10.2 Perturbations of the form $x + h(W(x))$

Consider

$$F_h(x) = x + h(W(x)), \quad (10.8)$$

where $h(y)$ has subexponential growth as $y \rightarrow +\infty$ and is regular in the relevant slow coefficient field. Put $y = W(x)$, so

$$z = ye^y + h(y). \quad (10.9)$$

Let $L = W(z)$, $q = e^{-L}$, and $y = L - v$. Dividing (10.9) by e^L and rearranging gives the exact implicit relation

$$B_L(v) = q \frac{h(L-v)}{L-v}, \quad B_L(v) = \frac{L}{L-v} - e^{-v}. \quad (10.10)$$

Once v is found as a q -series, the inverse is

$$F_h^{-1}(z) = z - h(L - v). \quad (10.11)$$

The leading correction follows immediately. Since $B'_L(0) = (L+1)/L$,

$$v = \frac{h(L)}{L+1} q + O(q^2), \quad (10.12)$$

and hence

$$F_h^{-1}(z) = z - h(L) + \frac{h(L)h'(L)}{L+1} q + O(q^2). \quad (10.13)$$

For $h(L) = \alpha L$, this reduces to (10.5). For a polynomial or polynomial-logarithmic h , every higher coefficient is obtained by the same triangular composition procedure and remains in the chosen slow field.

10.3 Ordinary near-identity perturbations

For a general

$$F(X) = X + H(X), \quad H(X) = o(X), \quad (10.14)$$

the direct formula (8.13) is usually sufficient. The special transform through W is advantageous only when the perturbation is naturally expressed in $W(X)$, because then the exact differential relation (8.6) and the identity $e^{-W(X)} = W(X)/X$ expose the correct transseries sectors without expanding nested logarithms prematurely.

11 Algorithms and symbolic implementation

11.1 Operator generation of the $1/z$ coefficients

The following short SymPy-style routine implements (4.6). It keeps every coefficient as an exact rational function of L .

Listing 1: Exact generation of the coefficient $C_n(L)$.

```
from sympy import symbols, diff, factorial, factor

L = symbols("L")

def D_block(k, R):
    """Return  $D_k R = L/(1+L) R' - k R$ ."""
    return L/(1 + L) * diff(R, L) - k*R

def C(n):
    """Coefficient of  $z^{*(-n)}$  in the inverse of  $x + W(x)$ ."""
    R = L**(n + 1)
    for k in range(n):
        #  $D_0, D_1, \dots, D_{n-1}$ 
        R = D_block(k, R)
    return factor((-1)**(n + 1) * R / factorial(n + 1))

for n in range(1, 7):
    print(n, C(n))
```

The approximation itself is

Listing 2: Evaluation of a sector truncation.

```
from mpmath import mp

mp.dps = 80

def inverse_approx(z, N, C_functions):
    Lval = mp.lambertw(z) # principal real branch
    ans = z - Lval
    for n in range(1, N + 1):
        ans += C_functions[n](Lval) / z**n
    return ans
```

11.2 Direct q -series reversion

A generic coefficient engine for (3.6) can instead operate on truncated ordinary power series.

Listing 3: Triangular reversion of $q = B(v, u)$.

```

from sympy import symbols, exp, series, solve, expand

q, u, v = symbols("q u v")
B = 1/(1 - u*v) - exp(-v)
V = 0

for n in range(1, N + 1):
    an = symbols(f"a{n}")
    trial = V + an*q**n
    residual = series(B.subs(v, trial) - q, q, 0, n + 1).remove0()
    equation = expand(residual).coeff(q, n)
    an_value = solve(equation, an)[0]
    V = expand(V + an_value*q**n)

```

This approach is preferable when the implicit relation is modified, as in (10.10). The operator method is preferable for the original near-identity problem because it avoids repeated series composition.

11.3 Hybrid workflow for a general logarithmic transseries

Recommended reversal workflow

1. Determine the dominant monomial and invert it exactly or asymptotically. Fix real or complex branches at this stage.
2. Compute the images of X^{-1} , $\log X$, and all deeper generators under the proposed leading inverse. Promote the coefficient field when a new scale appears.
3. Normalize to $F = X + H$ in a complete outer block algebra.
4. Compute two or three blocks by direct coefficient cancellation. This tests the scale choice and exposes any missing generators.
5. Switch to Newton reversion when many blocks are required, doubling the working cutoff after each update.
6. Expand adapted slow coordinates, such as $L = W(X)$, into ordinary iterated logarithms only after the exact outer coefficients are known.
7. Verify left, right, and derivative residuals. For actual functions, convert the composition residual into an error bound using a lower bound for $|F'|$.

11.4 Complexity and expression control

Exact arithmetic is strongly recommended. Coefficients such as (4.13) and (4.14) contain large cancellations that floating point arithmetic can obscure. Expression growth is controlled by:

- retaining factored rational functions until coefficients must be compared;
- using Horner evaluation for slow-variable polynomials;
- caching powers and derivatives;
- truncating after every multiplication, composition, and division;
- separating algebraic simplification from branch-dependent analytic simplification;
- representing the two cutoffs explicitly: outer sector order and inner inverse-log order.

12 Conclusions

The inverse of $x + W(x)$ has more structure than a single logarithmic asymptotic series suggests. The exact substitution $y = W(x)$ identifies it as

$$f^{-1}(z) = z - \mathcal{W}_1(z).$$

The further substitution $L = W(z)$, followed by $\mathcal{W}_1(z) = L - v$, converts the large- z problem into the local analytic reversion

$$\frac{L}{L - v} - e^{-v} = e^{-L}.$$

This yields a convergent expansion in the natural sector parameter $q = e^{-L} = L/z$, with rational coefficient functions and a simple Lagrange formula. The same coefficients arise from Lagrange–Bürmann inversion at infinity through the differential operators $\mathcal{D}_k = L(1 + L)^{-1} d/dL - k$.

The ordinary inverse-logarithmic expansion is obtained by replacing $L = W(z)$ by its Stirling-number series in $\log z$ and $\log \log z$. That sector alone is incomplete: the first correction to $z - W(z)$ is of order $W(z)/z$, smaller than every fixed inverse power of $\log z$. Keeping the outer $1/z$ sectors separate from the slow coefficient expansions resolves the hierarchy cleanly.

The reusable lesson is methodological. Series reversal of logarithmic transseries should proceed by dominant-scale normalization, generator-first composition, work in a slow differential coefficient field, and near-identity inversion by triangular recursion, Lagrange–Bürmann, or Newton iteration. Composition residuals are not merely formal checks; under a derivative lower bound they provide rigorous error certificates.

A Additional exact coefficients

With

$$A_n(L) = \frac{LR_n(L)}{n!(1 + L)^{2n-1}}, \quad C_n(L) = L^n A_n(L), \quad (\text{A.1})$$

the first six numerator polynomials are

$$\begin{aligned} R_1(L) &= 1, \\ R_2(L) &= L^2 - 2, \\ R_3(L) &= 2L^4 - L^3 - 12L^2 - 6L + 6, \\ R_4(L) &= 6L^6 - 8L^5 - 69L^4 - 40L^3 + 96L^2 + 72L - 24, \\ R_5(L) &= 24L^8 - 58L^7 - 428L^6 - 151L^5 + 1320L^4 + 1380L^3 - 480L^2 - 720L + 120, \\ R_6(L) &= 120L^{10} - 444L^9 - 2920L^8 + 424L^7 + 16577L^6 + 17892L^5 \\ &\quad - 13530L^4 - 26400L^3 - 1800L^2 + 7200L - 720. \end{aligned}$$

Thus

$$A_6(L) = \frac{LR_6(L)}{720(1 + L)^{11}}, \quad C_6(L) = \frac{L^7 R_6(L)}{720(1 + L)^{11}}. \quad (\text{A.2})$$

The highest-degree coefficient of R_n is $(n - 1)!$, while the constant term for the displayed cases is $(-1)^{n-1}n!$. The operator formula (4.6) is preferable to storing these expanded polynomials once n becomes large.

For comparison, the same coefficients in $u = 1/L$ begin as

$$\begin{aligned} A_1(u) &= \frac{1}{1+u}, \\ A_2(u) &= \frac{1-2u^2}{2(1+u)^3}, \\ A_3(u) &= \frac{(u-2)(6u^3+6u^2-1)}{6(1+u)^5}, \\ A_4(u) &= -\frac{24u^6-72u^5-96u^4+40u^3+69u^2+8u-6}{24(1+u)^7}, \\ A_5(u) &= \frac{120u^8-720u^7-480u^6+1380u^5+1320u^4-151u^3-428u^2-58u+24}{120(1+u)^9}. \end{aligned}$$

The limits $A_n(0) = 1/n$ are transparent in this coordinate.

B Derivation of the joint $(q, 1/L)$ expansion

This appendix records the calculation behind (3.23). Set

$$v = t + uv_1 + u^2v_2 + u^3v_3 + \cdots, \quad t = -\log(1-q), \quad s = 1-q. \quad (\text{B.1})$$

Since $e^{-t} = s$, substitution into

$$q = \frac{1}{1-uv} - e^{-v}$$

and comparison of powers of u gives

$$v_1 = -\frac{t}{s}, \quad (\text{B.2})$$

$$v_2 = \frac{t(2 + (2q-1)t)}{2s^2}, \quad (\text{B.3})$$

$$v_3 = -\frac{t}{6s^3} [6 + (18q-9)t + (2-6q+6q^2)t^2]. \quad (\text{B.4})$$

Therefore

$$\begin{aligned} V(q, u) &= t - \frac{ut}{s} + \frac{u^2t(2 + (2q-1)t)}{2s^2} \\ &\quad - \frac{u^3t}{6s^3} [6 + (18q-9)t + (2-6q+6q^2)t^2] + O(u^4). \end{aligned} \quad (\text{B.5})$$

At $q = 0$, every term vanishes, as required. Expanding (B.5) again in powers of q reproduces the large- L expansions of all $A_n(L)$ simultaneously.

C Stirling polynomials for the Lambert expansion

The polynomials P_n defined in (5.2) through order eight are

$$\begin{aligned} P_1(s) &= s, \\ P_2(s) &= \frac{s(s-2)}{2}, \end{aligned}$$

$$\begin{aligned}
P_3(s) &= \frac{s(2s^2 - 9s + 6)}{6}, \\
P_4(s) &= \frac{s(3s^3 - 22s^2 + 36s - 12)}{12}, \\
P_5(s) &= \frac{s(12s^4 - 125s^3 + 350s^2 - 300s + 60)}{60}, \\
P_6(s) &= \frac{s(20s^5 - 274s^4 + 1125s^3 - 1700s^2 + 900s - 120)}{120}, \\
P_7(s) &= \frac{s}{840} (120s^6 - 2058s^5 + 11368s^4 - 25725s^3 \\
&\quad + 24500s^2 - 8820s + 840), \\
P_8(s) &= \frac{s}{2520} (315s^7 - 6534s^6 + 45962s^5 - 142149s^4 \\
&\quad + 205800s^3 - 135240s^2 + 35280s - 2520).
\end{aligned}$$

They provide an arbitrary-order route from the exact W -coordinate coefficients $C_n(W(z))$ to polynomials in $\log z$ and $\log \log z$.

D A compact proof of the leading power-log inverse

For completeness, we verify (9.2). Let

$$Y = cX^a(\log X)^b, \quad b \neq 0,$$

and put $s = \log X$. Then

$$\left(\frac{Y}{c}\right)^{1/b} = se^{(a/b)s}.$$

Multiplication by a/b gives

$$\frac{a}{b} \left(\frac{Y}{c}\right)^{1/b} = \frac{a}{b} s \exp\left(\frac{a}{b}s\right).$$

Applying a suitable branch of W ,

$$\frac{a}{b}s = W_k\left(\frac{a}{b} \left(\frac{Y}{c}\right)^{1/b}\right),$$

so $X = e^s$ is exactly (9.2). This calculation illustrates why Lambert W is a natural normalizer for series reversal with a power-logarithmic leading monomial.

E Reference identities and sanity checks

The following identities are useful for independent verification:

$$Le^L = z, \quad e^{-L} = \frac{L}{z}, \quad L' = \frac{L}{z(1+L)}, \quad (\text{E.1})$$

$$\mathcal{W}_1(z)e^{\mathcal{W}_1(z)} + \mathcal{W}_1(z) = z, \quad g(z) = z - \mathcal{W}_1(z), \quad W(g(z)) = \mathcal{W}_1(z), \quad (\text{E.2})$$

$$g(z) + W(g(z)) = z, \quad g'(z) = \frac{1}{1+W'(g(z))}, \quad z - W(z) \leq g(z) \leq z. \quad (\text{E.3})$$

For symbolic coefficients, three equivalent checks are available:

1. compare A_n from ordinary Lagrange inversion with C_n/L^n from the differential operator;
2. substitute the truncated v -series into $B_L(v) - q$ and verify cancellation;
3. substitute G_N into $G_N + W(G_N) - z$ and verify the predicted residual order.

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